Contact Reconstruction and Spectral Balance for Surface Singularities with a Multiple Smooth Tangent Cone

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Abstract

We recover the degrees and all contact orders of surface germs $f = g_e^r + h_{er+1}$, $e \geq 3$, $r \geq 2$, where $V(g_e) \subset \mathbf{P}^2$ is smooth and $V(h_{er+1})$ is smooth at its finite intersection with $V(g_e)$. The monodromy characteristic polynomial yields these data by coefficient extraction and finite Möbius inversion; the local topological zeta function yields them through its poles and residues. Thus either invariant determines the other and classifies semisimplified Milnor monodromy in this family. We also obtain the spectral balance $\mu + M_r - M_1 = (er)^3 - (er)^2 + er$, where M_m is the multiplicity of an individual primitive mth root on H^2. At fixed e, r, it forces every monodromy-equivariant injection to be an isomorphism. We realize distinct contact profiles with equal Milnor numbers and incomparable monodromy, and describe the spectral change in a contact collision. Explicit formulas for the two invariants also yield simple actual poles, eigenvalues at the origin, and twisted topological holomorphy.

Keywords. Surface singularities; monodromy; topological zeta functions; tangency; contact reconstruction.

1 Introduction

We study how finite tangency affects the monodromy of surface germs with a multiple smooth tangent cone. The main result recovers both degrees and all contact orders from either the monodromy characteristic polynomial or the local topological zeta function. A spectral balance then governs comparisons between the monodromy representations of different profiles.

We work over $\mathbf{C}$. Let $g_e, h_D \in \mathbf{C}[x, y, z]$ be nonzero homogeneous polynomials of degrees e, D, respectively, and set

$$f = g_e^r + h_D, \qquad e \geq 3, \quad r \geq 2, \quad A = er, \quad D = A + 1. \tag{1.1}$$

Assume that the zero scheme $C = V(g_e) \subset \mathbf{P}^2$ is smooth, that $\mathcal{P} = C \cap H$ is finite for $H = V(h_D)$, and that H is smooth at every point of $\mathcal{P}$. For $p \in \mathcal{P}$, put $q_p = I_p(C, H)$. The finite multiset $\{q_p\}$ is the *contact profile*. These germs have isolated critical points, as proved in Lemma 2.1.

Write $Z_f(s) = Z_{\mathrm{top},0}(f; s)$ for the local topological zeta function with standard volume form. Let F_f be the Milnor fiber at the origin, T_f its monodromy on $H^2(F_f, \mathbf{C})$, and

$$\Delta_f(t) = \det(1 - tT_f \mid H^2(F_f, \mathbf{C})).$$

Write μ_f for its Milnor number. Our first result reconstructs the entire profile from either one of these invariants.

Theorem 1.1 (Reconstruction)**.** *Each of Δ_f and Z_f uniquely determines $(e, r, \{q_p\})$ in the family (1.1). For two members, equality of these data is equivalent to equality of either invariant and to isomorphism of their semisimplified Milnor monodromy. Recovery from Δ_f uses coefficient extraction and finite Möbius inversion.*

For Δ_f, the initial step is visible in a single term. If $P_f = (1-t)\Delta_f$ begins as $1-\kappa t^A+\cdots$, then

$$A = er, \qquad \kappa = e^2 - 3e + 3, \qquad e = \frac{3+\sqrt{4\kappa-3}}{2}, \qquad r = \frac{A}{e}. \tag{1.2}$$

Two further coefficients recover the number of intersection points and the number of contacts of order at least two. The logarithmic derivative of the remaining contact polynomial then recovers each order and its frequency. For Z_f, the largest actual pole and its residue first recover e, r; the remaining contact poles and their residues recover the higher contact orders and their frequencies; Bézout gives the number of transverse intersections (Proposition 5.4). The recovered data also determine every twisted topological zeta function (Corollary 7.4).

A collision of q transverse intersections into one contact of order q increases the Milnor number by $q-1$. At the same time it decreases the multiplicity of every primitive rth root of unity by $q-1$. Our second result expresses this compensation as a balance valid throughout the family. Let $M_m(f)$ denote the algebraic multiplicity of an individual primitive mth root as an eigenvalue on $H^2(F_f, \mathbf{C})$.

Theorem 1.2 (Spectral balance and incomparability)**.** *For every member of (1.1),*

$$\mu_f + M_r(f) - M_1(f) = A^3 - A^2 + A. \tag{1.3}$$

For two members with the same e, r, every monodromy-equivariant injection $H^2(F_f, \mathbf{C}) \hookrightarrow H^2(F_{\widetilde{f}}, \mathbf{C})$ is an isomorphism. If their contact profiles differ, neither direction admits such an injection, also after semisimplification.

This incomparability occurs even at equal Milnor number. For $e = 4, r = 2$, the profiles $\{2, 4, 1^{30}\}$ and $\{3, 3, 1^{30}\}$ both occur and both give $\mu = 419$. The first has primitive 20th eigenvalues and the second primitive 30th eigenvalues, with neither order present in the other (Example 6.1).

Both reconstruction procedures start from the same calculation. Blowing up the origin gives three local models, over $\mathbf{P}^2 \setminus C$, $C \setminus \mathcal{P}$, and $\mathcal{P}$. Their weighted Newton and monodromy formulas assemble into a polynomial contact transform for Δ_f and a rational one for Z_f.

The resolution formula for the topological zeta function is due to Denef and Loeser [3], and the monodromy computation uses A'Campo's formula [1]. The partial-resolution formula of Gusein-Zade, Luengo, and Melle-Hernández [4, Theorem 1 and Corollary 1] provides the multiplicative assembly; their §3.5 includes the transverse specialization of our monodromy formula. Melle-Hernández's formula, recorded in [4, §3.4, Remark 2], gives the Milnor number for arbitrary contacts. Building on these forward calculations, we recover the full contact data from either invariant and derive the spectral comparison.

Related monodromy results concern Newton nondegenerate surfaces [6], their local motivic zeta functions [5, Theorem 7.2.1], and generalized suspensions and Lê–Yomdin surfaces [2]. The present tangent cone is the non-reduced divisor rC; the surface Lê–Yomdin class has a reduced tangent cone [2, Definition 7.1]. Likewise, the lowest-degree face polynomial g_e^r and all its first derivatives vanish on the nonempty torus part of C, while the three models after blowing up the origin are Newton nondegenerate. For a broader account of the monodromy conjecture, see [8].

Sections 2–4 compute the local models and global invariants. Section 5 inverts the two contact transforms, placing the pole calculation alongside the reconstruction from Z_f. Section 6 proves spectral balance and treats contact collisions and extremal members of the

family. Section 7 gives the twisted formulas and holomorphy. The appendices contain the cone calculations and a worked example of reconstruction.

2 Three local models after blowing up the origin

2.1 Contact data

Write

$$c = \chi(C) = e(3-e), \qquad \kappa = 3 - c, \qquad n = eD = \sum_{p\in\mathcal{P}} q_p, \qquad b = |\mathcal{P}|,$$
$$n_q = |\{p : q_p = q\}|, \qquad S = \sum_{q\geq 2}(q-1)n_q = n - b, \tag{2.1}$$
$$B = (r-1)D + 1, \qquad \beta = 3r - 2.$$

Here χ denotes Euler characteristic with compact supports; $c \leq 0$ and $b \geq 1$.

2.2 Resolution conventions and isolatedness

Monodromy is taken along a counterclockwise loop in the base. For a log resolution $h : Y \to \mathbf{C}^3$, let E_i be the components of the total transform of $f^{-1}(0)$ and write

$$\operatorname{div}(f \circ h) = \sum_i N_i E_i, \qquad \operatorname{div}(h^*(dx \wedge dy \wedge dz)) = \sum_i (\nu_i - 1)E_i.$$

For a nonempty index set I, put $E_I^\circ = (\bigcap_{i\in I} E_i) \setminus \bigcup_{j\notin I} E_j$. Our normalization is

$$Z_f(s) = \sum_{\varnothing\neq I} \chi(E_I^\circ \cap h^{-1}(0)) \prod_{i\in I} (N_i s + \nu_i)^{-1}. \tag{2.2}$$

For a local function F, let ω be a nonzero holomorphic top-degree form with $\operatorname{Supp}(\operatorname{div}\omega) \subset F^{-1}(0)$. On a common log resolution of F and ω, taken to be an isomorphism outside $F^{-1}(0)$, use (2.2) with $\nu_i - 1 = \operatorname{ord}_{E_i}(h^*\omega)$. The form may vanish along $F^{-1}(0)$.

Auxiliary coordinate boundaries with data $(N, \nu) = (0, 1)$ may be used to refine the Euler strata. They are additional to the components of the function divisor: their denominator factor is 1, so Euler additivity preserves the sum.

For any local Milnor fiber we take

$$\zeta(t) = \prod_j \det(1 - tT_j \mid H^j)^{(-1)^{j+1}}. \tag{2.3}$$

This is the reciprocal of the convention in [4]. In this convention A'Campo's formula reads

$$\zeta(t) = \prod_{i:N_i>0} (1 - t^{N_i})^{-\chi(E_i^\circ \cap h^{-1}(0))}. \tag{2.4}$$

In (2.4), E_i° removes all other positive multiplicity components, including the strict transform.

Lemma 2.1. *The function f in (1.1) is reduced and has an isolated critical point at the origin. Its Milnor fiber is connected, with reduced cohomology concentrated in degree 2. In particular,*

$$\zeta_f(t) = \big((1-t)\Delta_f(t)\big)^{-1}. \tag{2.5}$$

Proof. At a nonzero point with $f = df = 0$, the homogeneous Euler identity and the equation $f = 0$ give

$$Ag_e^r + Dh_D = 0, \qquad g_e^r + h_D = 0.$$

Since $D - A = 1$, both g_e and h_D vanish. The corresponding projective point lies in $\mathcal{P}$, where $dh_D \neq 0$. But $r \geq 2$ gives $df = dh_D$ at that point, a contradiction.

If the critical locus had a positive dimensional germ through the origin, it would contain an analytic curve through the origin. The restriction of f to that curve would be constant, hence zero, contrary to what was just proved. A repeated factor of f would also give a positive dimensional critical locus. The Milnor bouquet theorem [7] now gives the cohomology statement and (2.5). □

2.3 The local equations

Let $\pi_0 : X_1 \to \mathbf{C}^3$ be the blowup of the origin and let $E = \pi_0^{-1}(0) \simeq \mathbf{P}^2$. Near C choose u normal to E, v an equation for C in E, and w along C. Then

$$f \circ \pi_0 = u^A(v^r + uh(v,w)), \qquad \pi_0^*(dx \wedge dy \wedge dz) = u^2 \cdot \text{unit}\, du \wedge dv \wedge dw. \tag{2.6}$$

Outside C the function is u^A times a unit. On $C \setminus \mathcal{P}$, replacing u by $h(v,w)u$ gives

$$G = u^A(v^r + u).$$

At a point of contact order $q \geq 2$, smoothness of H implies $\partial_v h(0) \neq 0$. The implicit function theorem gives $h = \alpha(v,w)(v + \phi(w))$, where α is a unit and $\operatorname{ord}_0 \phi = q$. A change of the coordinate w makes $\phi(w) = w^q$, and the change $u' = \alpha u$ gives

$$\mathcal{T}_q = u^A(v^r + uv + uw^q). \tag{2.7}$$

When $q = 1$, take $w' = h - v$ to obtain the same expression. These changes multiply the function and the top-degree form by units. Their orders and divisor strata on a common resolution are therefore unchanged. The three local models are computed with the form

$$\omega = u^2\, du \wedge dv \wedge dw.$$

2.4 Euler assembly

The three local computations assemble by Euler integration. For monodromy this is the partial-resolution formula of [4, Theorem 1 and Corollary 1].

Lemma 2.2. *Let Z_G, Z_q be the local topological functions of $G, \mathcal{T}_q$ with the form ω, and let ζ_G, ζ_q be their monodromy zeta functions. Then*

$$Z_f = \kappa Z_{u^A} + (c-b)Z_G + \sum_{q \geq 1} n_q Z_q, \tag{2.8}$$

$$\zeta_f = \zeta_{u^A}^{\kappa} \zeta_G^{c-b} \prod_{q \geq 1} \zeta_q^{n_q}. \tag{2.9}$$

Proof. Choose a log resolution $\rho : Y \to X_1$ of the pulled-back function and form, and set $h = \pi_0 \rho$. The pulled-back form on Y is $h^*(dx \wedge dy \wedge dz)$, so its numerical data coincide with those of the original germ. For each resolution stratum, Euler Fubini gives

$$\chi(E_I^\circ \cap h^{-1}(0)) = \int_E \chi(E_I^\circ \cap \rho^{-1}(y))\, d\chi(y).$$

Multiplying by its fixed denominator and summing yields the Euler integral of the local weighted topological function. Local changes of variables preserve it by [2, Proposition 1.3]. The corresponding normal crossings identity will also be used for the twisted functions. Blowing up a smooth center of codimension v having normal crossings with the boundary and contained in exactly k of its components replaces their linear denominators L_i by an additional

denominator $L_* = \sum_{i=1}^{k} L_i + v - k$. The nonzero Euler strata of the projective exceptional fiber give

$$\frac{1}{L_*}\left[\sum_{i=1}^{k}\prod_{j\neq i} L_j^{-1} + (v-k)\prod_{i=1}^{k} L_i^{-1}\right] = \prod_{i=1}^{k} L_i^{-1}. \tag{2.10}$$

Boundary components transverse to the center contribute common factors on the two sides. The identity therefore applies with the orders of the pulled-back form.

For monodromy, on this fixed resolution put

$$a_N(y) = \sum_{i:N_i=N} \chi(E_i^\circ \cap \rho^{-1}(y)).$$

The local A'Campo expression is $\zeta_{f\circ\pi_0,y} = \prod_N (1-t^N)^{-a_N(y)}$. Euler Fubini identifies the global exponent with $-\int_E a_N\, d\chi$ for each N, giving the multiplicative Euler integral of these local expressions. Finally stratify

$$E = (\mathbf{P}^2 \setminus C) \sqcup (C \setminus \mathcal{P}) \sqcup \bigsqcup_{p\in\mathcal{P}} \{p\},$$

whose Euler characteristics are $\kappa, c-b, 1$. The local resolution invariants are constant on these strata, giving (2.8) and (2.9). □

3 The local tangency calculation

Use the linear functions

$$a = As+3, \quad Q = rDs+3r+1, \quad d = s+1, \quad J = Ds+4, \quad K_q = qBs+q\beta+1. \tag{3.1}$$

Three identities used below are

$$Q = ra+rd-(r-1) = rJ-(r-1), \qquad K_q = q(Q-a)+1, \qquad B-(r-1)D = 1. \tag{3.2}$$

Proposition 3.1 (Local contact formula)**.** *For the local models in Section 2,*

$$Z_{u^A} = \frac{1}{a}, \qquad \zeta_{u^A} = (1-t^A)^{-1}, \tag{3.3}$$

$$Z_G = \frac{r-1+a^{-1}+d^{-1}}{Q}, \qquad \zeta_G = \frac{1-t^{rD}}{1-t^D}, \tag{3.4}$$

$$Z_q = \frac{a^{-1}+r/d}{Q} - \frac{(q-1)rsJ}{dQK_q}, \qquad \zeta_q = \frac{1-t^B}{1-t^{qB}}. \tag{3.5}$$

Proof. The topological function of $\mathcal{T}_q$. The Newton vertices of $\mathcal{T}_q$ are

$$X = (A, r, 0), \qquad Y = (D, 1, 0), \qquad Z = (D, 0, q).$$

The primitive facet normals and their numerical data for $\omega = u^2\,du\wedge dv\wedge dw$ are

Normal	N	ν	Denominator
$\mathsf{U} = (1,0,0)$	A	3	a
$\mathsf{V} = (0,1,0)$	0	1	1
$\mathsf{W} = (0,0,1)$	0	1	1
$\mathsf{R} = (r,1,0)$	rD	$3r+1$	Q
$\mathsf{S} = (0,q,1)$	q	$q+1$	$p_q = qs+q+1$
$\mathsf{T} = (q(r-1),q,1)$	qB	$q\beta+1$	K_q

All compact edges have normalized length one. The cross product of $Y-X$ and $Z-X$ is $-\mathsf{T}$, so the compact triangle has normalized area one. The edge polynomials are primitive binomials, and the triangle polynomial has nonzero w-derivative on the torus. They are nondegenerate. A regular subdivision of the Newton fan consequently gives a toric log resolution near the fiber over the origin.

The weighted orbit calculation in Appendix A gives

$$Z_q = \frac{1}{K_q}\left[\frac{q}{a} + \frac{1}{aQ} + \frac{r-1}{Q} - (q-1) + \frac{qr - 1 + Q^{-1}}{d}\right]. \tag{3.6}$$

Using (3.2), its numerator satisfies

$$adQK_qZ_q = K_q(d+ra) - (q-1)ra(d-1)(a+d).$$

Since $d-1=s$ and $a+d=J$, this proves the first formula in (3.5).

The monodromy of $\mathcal{T}_q$. In the cone dual to XY, insert the rays

$$Q_j = (j(r-1), j, 1), \qquad 0 \le j \le q.$$

Here $Q_0 = \mathsf{W}$, $Q_q = \mathsf{T}$, and the cross product of consecutive vectors is $(-1, r-1, 0)$. These two-dimensional cones are regular and can be kept in a regular refinement of the full fan.

In A'Campo's formula, the two-dimensional torus orbit of E_{T} has complement of Euler characteristic $+1$: the strict transform is $1+\xi+\eta=0$ in $(\mathbf{C}^*)^2$. An orbit of a two-dimensional cone can remain in a single positive multiplicity divisor only if its other ray has $N=0$. This forces $\rho_u = 0$ and $\rho_v\rho_w = 0$, so the only such rays are V, W. The cones containing V give Euler characteristic zero. Among the cones containing W, the sole nonzero contribution comes from the edge XY. It is $\mathbf{C}^*$ minus one point, has Euler characteristic -1, and belongs to the divisor E_{Q_1}, of multiplicity B.

There is no three-dimensional cone with just one positive multiplicity ray: that would require both V and W, which lie in no common Newton cone. All other orbits have an extra torus factor and Euler characteristic zero. Also, the strict transform's own open stratum misses the fiber, since every orbit over the origin lies in a positive multiplicity toric divisor. Thus the divisor Euler numbers are $+1$ at T and -1 at Q_1. For $q=1$ these are the same divisor and the contributions cancel. These cases exhaust the fiber orbits relevant to A'Campo's formula in every regular refinement preserving the indicated two-dimensional cones. Formula (2.4) now gives ζ_q.

The other two models. For G, take a plane resolution in u, v and its product with the smooth w-direction. The Newton vertices are (A, r) and $(D, 0)$, with normals $(1,0), (r,1), (0,1)$ and denominators $a, Q, 1$. The two vertex cones have lattice indices $1, r$, giving $1/(aQ)$ and r/Q; the primitive edge gives $-s/(dQ)$ by the orbit computation. Their sum is (3.4). For its monodromy insert the rays $(j,1)$, $1 \le j \le r$, between $(0,1)$ and $(r,1)$. The divisor at $(r,1)$ has open Euler characteristic -1, and the divisor at $(1,1)$ has an additional point of Euler characteristic $+1$ next to the zero multiplicity boundary. Their multiplicities are rD and D. This gives ζ_G. The monomial formulas follow immediately. □

4 Contact transforms and global invariants

4.1 The two contact transforms

The local formulas express the global invariants through the contact polynomial and the rational contact transform:

$$\Gamma_f(u) = \prod_{q\ge 2}(1 + u + \cdots + u^{q-1})^{n_q}, \qquad U_f(x) = \sum_{q\ge 2}\frac{n_q(q-1)}{1+qx}. \tag{4.1}$$

The total contact excess is $\deg \Gamma_f = U_f(0) = S$.

Theorem 4.1 (Contact formulas)**.** *The global invariants satisfy*

$$Z_f(s) = \frac{\kappa}{a} + \frac{(c-b)(r-1)}{Q} + \frac{c}{aQ} + \frac{c+b(r-1)}{dQ} - \frac{rsJ}{dQ} U_f(Bs+\beta), \tag{4.2}$$

$$\Delta_f(t) = \frac{(1-t^A)^\kappa}{1-t} \left(\frac{1-t^{rD}}{1-t^D}\right)^{b-c} \Gamma_f(t^B). \tag{4.3}$$

In particular,

$$\begin{aligned} \mu_f &= A\kappa + (r-1)D(b-c) + B(n-b) - 1 \\ &= A^3 - A^2 + A - 1 + c - b. \end{aligned} \tag{4.4}$$

Proof. Insert Proposition 3.1 into Lemma 2.2. For the topological function, the coefficients of $1/(aQ)$ combine to c, and those of $1/(dQ)$ combine to $c + b(r-1)$. The remaining contact sum is exactly $U_f(Bs+\beta)$. For monodromy, (2.9) gives

$$\zeta_f = (1-t^A)^{-\kappa} \left(\frac{1-t^{rD}}{1-t^D}\right)^{c-b} \prod_{q\geq 2} \left(\frac{1-t^B}{1-t^{qB}}\right)^{n_q}.$$

Use (2.5) to obtain (4.3). Its degree is (4.4). Polynomiality is also visible directly: the geometric-series quotients are polynomials, and $(1-t^A)^\kappa/(1-t)$ is a polynomial. Using $n = eD$ and $B - (r-1)D = 1$ simplifies the degree formula to its second expression. □

The classical formula of Melle-Hernández [4, §3.4, Remark 2] gives the same Milnor number: $\operatorname{Sing} V(g_e^r) = C$ is disjoint from $\operatorname{Sing} H$, and $I_p(V(g_e^r), H) = rq_p$. Taking $d = A$, $k = 1$, and $\sum_p q_p = eD$ gives (4.4).

Let $Z_{\mathrm{tr}}, \Delta_{\mathrm{tr}}, \mu_{\mathrm{tr}}$ denote the specialization in which all $n = eD$ intersections are transverse. The formulas for $\zeta_{\mathrm{tr}}^{-1} = (1-t)\Delta_{\mathrm{tr}}$ and μ_{tr} agree with [4, §3.5], taking their $d = A$, $k = 1$, and a single smooth component of degree e and multiplicity r. Substituting $b = n - S$ in the preceding theorem gives the compact form of the contact contribution:

$$Z_f - Z_{\mathrm{tr}} = \frac{s}{(s+1)Q(s)} \left[(r-1)U_f(0) - r(Ds+4)U_f(Bs+\beta)\right], \tag{4.5}$$

$$\frac{\Delta_f(t)}{\Delta_{\mathrm{tr}}(t)} = \left(\frac{1-t^D}{1-t^{rD}}\right)^S \Gamma_f(t^B), \qquad \mu_f - \mu_{\mathrm{tr}} = S. \tag{4.6}$$

Thus a contact changes both the contribution along the curve and the local term at the intersection point. The two terms in (4.5) record these changes separately. The function U_f has positive simple-pole weights; for $x > 0$ and $k \geq 0$,

$$(-1)^k U_f^{(k)}(x) = k! \sum_{q\geq 2} \frac{n_q (q-1) q^k}{(1+qx)^{k+1}} \geq 0. \tag{4.7}$$

4.2 Characters of Milnor monodromy

To read individual eigenvalues, take a common cyclic group C_M, where $M = \operatorname{lcm}(A, rD, \{qB : n_q > 0,\ q \geq 2\})$. For $N \mid M$, let Reg_N be the regular representation of C_N pulled back to C_M. For $qN \mid M$, set

$$W_{q,N} = \ker(\mathrm{Reg}_{qN} \longrightarrow \mathrm{Reg}_N),$$

where the map sends the basis vector indexed by j to the one indexed by $j \bmod N$. A character of order m has multiplicity $\mathbf{1}_{m|qN} - \mathbf{1}_{m|N}$ in $W_{q,N}$.

Proposition 4.2. *The semisimplification of Milnor monodromy has character*

$$[H^2(F_f, \mathbf{C})_{\mathrm{ss}}] = \kappa[\mathrm{Reg}_A] - [\mathbf{1}] + (b-c)[W_{r,D}] + \sum_{q\geq 2} n_q[W_{q,B}]. \tag{4.8}$$

For every $m \geq 1$ its eigenvalue multiplicity is

$$\begin{aligned} M_m(f) = \kappa \mathbf{1}_{m|A} - \mathbf{1}_{m=1} + (b-c)(\mathbf{1}_{m|rD} - \mathbf{1}_{m|D}) \\ + \sum_{q\geq 2} n_q(\mathbf{1}_{m|qB} - \mathbf{1}_{m|B}). \end{aligned} \tag{4.9}$$

In particular,

$$M_1(f) = 2 - c = 2g(C), \qquad M_r(f) = b + 3 - 2c. \tag{4.10}$$

Proof. All roots of (4.3) are Mth roots of unity, so the semisimple action factors through C_M. Taking characteristic polynomials gives (4.8), and then (4.9). This is a positive character:

$$\kappa\,\mathrm{Reg}_A - \mathbf{1} = (\kappa - 1)\,\mathrm{Reg}_A \oplus \mathrm{Aug}_A,$$

where Aug_A is the augmentation kernel. The remaining coefficients are nonnegative, since $b - c > 0$.

For $m = 1$ each W term has multiplicity zero and the first two terms give $\kappa - 1 = 2 - c$. For $m = r$, use $r \mid A$, $r \nmid D$, and $r \mid B$, the last identity following from $B = r(e(r-1)+1)$. This gives $M_r = \kappa + b - c = b + 3 - 2c$. □

5 Reconstructing the contact profile

5.1 Recovery from monodromy coefficients

The first step recovers the parameters before separating any contact factors. Put

$$P_f(t) = (1-t)\Delta_f(t).$$

Equation (4.3) reads

$$P_f(t) = (1-t^A)^\kappa (1 + t^D + \cdots + t^{(r-1)D})^{b-c} \Gamma_f(t^B). \tag{5.1}$$

Proposition 5.1 (Coefficient recovery)**.** *The coefficients of Δ_f determine*

$$\begin{aligned} A = \mathrm{ord}_{t=0}(P_f - 1), \qquad \kappa = -[t^A]P_f, \qquad e = \frac{3 + \sqrt{4\kappa - 3}}{2}, \qquad r = \frac{A}{e}, \\ b = c + [t^{A+1}]P_f, \qquad \theta := \sum_{q\geq 2} n_q = [t^B]P_f, \qquad n_1 = b - \theta. \end{aligned} \tag{5.2}$$

Proof. Since $A < D < B$, the first nonconstant term in (5.1) is $-\kappa t^A$. The identity $\kappa = e^2 - 3e + 3$ gives $4\kappa - 3 = (2e-3)^2$; $e \geq 3$ selects the displayed root. The coefficient of t^D comes only from the second factor, as $2A > D$ and $B > D$, and equals $b - c$.

To read the coefficient of t^B, consider the first two factors. Every degree in their product has the form $iA + jD$, with $i, j \geq 0$. If it equaled $B = (r-1)A + r$, then $j \leq r - 1$, whereas reduction modulo A would give $j \equiv r \bmod A$. This is impossible because $r < A$. Thus only the linear term of $\Gamma_f(t^B)$ contributes at degree B, giving $[t^B]P_f = \Gamma_f'(0) = \theta$. □

With e, r, b known, extract Γ_f from

$$\frac{P_f(t)}{(1-t^A)^\kappa} \left(\frac{1-t^D}{1-t^{rD}}\right)^{b-c} = \Gamma_f(t^B). \tag{5.3}$$

The substitution $u \mapsto t^B$ is injective, so this determines $\Gamma_f(u)$ uniquely. Its degree is $S = n - b$.

Proposition 5.2 (Finite inversion)**.** *Write* $\Gamma_f(u) = \sum_{j=0}^{S} \gamma_j u^j$*, with* $\gamma_0 = 1$*, and extend* $\gamma_j = 0$ *for* $j > S$*. Define*

$$L_m = m\gamma_m - \sum_{j=1}^{m-1} \gamma_j L_{m-j}. \tag{5.4}$$

Then the entire profile is recovered by

$$n_q = -\frac{1}{q} \sum_{d|q} \mu_{\mathrm{Mob}}(q/d) L_d \quad (q \geq 2), \qquad n_1 = b - \gamma_1. \tag{5.5}$$

It suffices to compute L_m *for* $m \leq S + 1$*.*

Proof. Recurrence (5.4) is coefficient comparison in $\Gamma_f(u) \sum_{m \geq 1} L_m u^m = u\Gamma_f'(u)$. On the other hand, the product in (4.1) gives

$$\frac{u\Gamma_f'(u)}{\Gamma_f(u)} = \sum_{q \geq 2} n_q \left(\frac{u}{1-u} - \frac{qu^q}{1-u^q} \right).$$

Consequently $L_m = \theta - \sum_{q|m,\ q \geq 2} q n_q$. For $q > 1$, Möbius inversion cancels the constant θ and gives (5.5). Finally, an occurring $q \geq 2$ satisfies $q - 1 \leq S$. If $S = 0$, then $\Gamma_f = 1$ and all intersections are transverse. □

Appendix B carries out this recovery for an integer-coefficient member of the family.

Lefschetz numbers. The multiplicity of the surface germ is also determined by its Lefschetz numbers. Let $\Lambda_m = 1 + \mathrm{Tr}(T_f^m \mid H^2(F_f, \mathbf{C}))$. The formal identity

$$-t\frac{P_f'(t)}{P_f(t)} = \sum_{m \geq 1} \Lambda_m t^m$$

and the initial term $-\kappa t^A$ give $\Lambda_m = 0$ for $1 \leq m < A$ and $\Lambda_A = A\kappa$. Thus A is the first index with a nonzero Lefschetz number.

5.2 Recovery from poles and residues

To reconstruct from Z_f, we locate its poles and compute their residues. The residue at the largest pole distinguishes the cubic case from degrees $e \geq 4$.

Proposition 5.3. *All actual poles of* Z_f *are simple. Its largest pole* σ_* *and corresponding residue* R_* *are*

$$(\sigma_*, R_*) = \begin{cases} (-3/A, -3(e-1)/A), & e \geq 4, \\ (-1/r, 4/r), & e = 3. \end{cases} \tag{5.6}$$

Every occurring $q \geq 2$ *gives a distinct actual pole* $\sigma_q = -(q\beta + 1)/(qB)$*, with*

$$\mathrm{Res}_{s=\sigma_q} Z_f = -\frac{n_q(q-1)r\sigma_q(D\sigma_q + 4)}{qB(\sigma_q + 1)Q(\sigma_q)} \neq 0. \tag{5.7}$$

If $e \geq 4$*, then* $\sigma_Q = -(3r+1)/(rD)$ *is also an actual pole with positive residue.*

Proof. Formula (4.2) gives the candidate list

$$-1, \quad -3/A, \quad -(3r+1)/(rD), \quad \sigma_q \quad (q \geq 2,\ n_q > 0). \tag{5.8}$$

Degrees $e \geq 4$. The strict inequalities

$$0 < \frac{3}{A} < \frac{3r+1}{rD} < \frac{\beta}{B} < \frac{q\beta+1}{qB} < 1 \tag{5.9}$$

show that the candidate roots are distinct, apart from repeated occurrences of the same contact order. The numerators of the first two differences are $r(e-3)$, and $B-\beta = (r-1)(D-3)$ gives the last inequality. Thus every actual pole is simple. At $s = -3/A$, $Q(s) = (e-3)/e$, so the residue is

$$\frac{\kappa + c/Q(-3/A)}{A} = -\frac{3(e-1)}{A}.$$

This proves the first case of (5.6).

The residue at a contact root is (5.7). All its factors are nonzero except possibly $D\sigma_q + 4$. If that factor vanished, then

$$D = q\big[(r-2)D + 4\big].$$

For $r \geq 3$ the right hand side is larger than D. For $r = 2$ this would say that the odd integer $D = 2e+1$ equals $4q$. Both are impossible.

At the root σ_Q of Q, set

$$\delta = \frac{e-3}{D}, \qquad \eta = \frac{(3r+1)(r-1)}{rD[rD-(3r+1)]}.$$

At σ_Q we have $Bs + \beta = \delta$ and $rJ = r-1$. The transverse residue, obtained from (4.2) with $b = n$, is $\eta e(2D+e-3)$. By (4.5), the contact correction has residue $-\eta[S - U_f(\delta)]$. Consequently

$$\operatorname{Res}_{s=\sigma_Q} Z_f = \eta\big[e(D+e-3) + b + U_f(\delta)\big] > 0. \tag{5.10}$$

The cubic case. Now $c = 0$, $B = r\beta$, and

$$a = 3(rs+1), \qquad Q = D(rs+1), \qquad D = 3r+1.$$

The sole term in (4.2) with denominator aQ has coefficient zero. More explicitly, the formula simplifies to

$$Z_f(s) = \frac{4}{rs+1} - \frac{3}{s+1} + \sum_{q\geq 2} n_q \frac{C_q s}{(s+1)K_q(s)}, \qquad C_q = \frac{(q-1)[q(r-1)\beta - D]}{D} > 0. \tag{5.11}$$

To obtain it, subtract the transverse formula and use $(r-1)K_q - rJ = [q(r-1)\beta - D](rs+1)$. For positivity, the smallest possible bracket is bounded below by $6r^2 - 13r + 3$, which equals 1 at $r = 2$ and increases for $r \geq 2$.

The contact roots are now $\sigma_q = -1/r - 1/(qB) \in (-1, -1/r)$ and $K_q(-1/r) = 1$. The common root $-1/r$ has residue $4/r$, while every contact residue has the simpler expression

$$\operatorname{Res}_{s=\sigma_q} Z_f = \frac{n_q C_q \sigma_q}{qB(\sigma_q + 1)} < 0. \tag{5.12}$$

Finally, using $\sum q n_q = 3D$, the residue at -1 is

$$\operatorname{Res}_{s=-1} Z_f = -\frac{1}{D}\left[b + 3r \sum_{q\geq 2} \frac{n_q(q-1)}{q(r-1)\beta - 1}\right] < 0. \tag{5.13}$$

This proves all assertions at the cubic boundary as well. □

The remaining candidate -1 has residue

$$\operatorname{Res}_{s=-1} Z_f = \frac{c + b(r-1) + r(D-4)\displaystyle\sum_{q\geq 2} \frac{n_q(q-1)}{q(r-1)(D-3)-1}}{3r+1-rD}. \tag{5.14}$$

Together with Proposition 5.3, this determines the pole set: -1 is present exactly when this residue is nonzero.

Proposition 5.4. *The rational function Z_f determines e, r and the contact profile.*

Proof. Take the largest actual pole σ_* and its residue R_*. Proposition 5.3 distinguishes the two parameter ranges by the sign of R_*:

$$\begin{cases} R_* > 0: & e = 3, \qquad r = -1/\sigma_*; \\ R_* < 0: & e = 1 + R_*/\sigma_*, \qquad r = -3/(e\sigma_*). \end{cases} \tag{5.15}$$

Now all the numerical constants in (3.1) are known. Remove the positions $-1, -3/A, -(3r+1)/(rD)$ from the pole set. Its remaining elements are exactly the occurring contact poles. At such a pole σ_q with residue R_q, use

$$q = -\frac{1}{B\sigma_q + \beta}, \qquad n_q = -\frac{qB(\sigma_q+1)Q(\sigma_q)}{(q-1)r\sigma_q(D\sigma_q+4)}\, R_q. \tag{5.16}$$

The denominators are nonzero by Proposition 5.3. Finally $n_1 = eD - \sum_{q\geq 2} q n_q$. □

Proof of Theorem 1.1. Propositions 5.1 and 5.2 recover all the data from Δ_f, and Proposition 5.4 recovers them from Z_f. Conversely, the contact formulas determine both invariants from those data. This proves the three equivalent conditions and the stated recovery procedures. Over $\mathbf{C}$, the characteristic polynomial determines a linear automorphism up to semisimplification, which proves the classification statement. □

5.3 Monodromy and analytic moduli

Corollary 5.5. *Every actual pole s_0 of Z_f gives the eigenvalue* $\exp(2\pi i s_0)$ *on the cohomology of F_f.*

Proof. Use the positive character (4.9). For $s_0 = -3/A$, the eigenvalue order is $m = A/\gcd(A,3) > 1$, detected by Reg_A. For $s_0 = -(3r+1)/(rD)$, it is $m = rD/\gcd(rD, 3r+1) > 1$. If $m \mid D$, then $rD \mid (3r+1)D$, which would force $r \mid 3r+1$. Thus $W_{r,D}$ detects this order. For a contact pole, the order is $m = qB/\gcd(q\beta+1, qB)$. If $m \mid B$, then $qB \mid (q\beta+1)B$, contradicting $q \nmid q\beta+1$. It is therefore detected by $W_{q,B}$. The order at -1 is 1, with multiplicity $M_1 = 2 - c > 0$. These cases also cover the common old root when $e = 3$. All primitive roots of each detected order occur with the same multiplicity by (4.9). □

Example 5.6 (Different curves with the same invariants)**.** Take two smooth plane cubics with different j-invariants, fix the same $r \geq 2$, and choose a degree $3r+1$ curve transverse to each. Both profiles consist of $3(3r+1)$ ones, so the corresponding Z_f and Δ_f agree. The resulting germs are not analytically contact equivalent. Such an equivalence would identify their lowest homogeneous terms up to a linear change of coordinates and a nonzero scalar. Their tangent-cone supports would then be projectively equivalent cubics, contrary to their distinct j-invariants.

6 Spectral balance and contact collisions

6.1 The balance identity

Proof of Theorem 1.2. Equation (4.10) gives $M_r - M_1 = b + 1 - c$. Adding this to the second expression in (4.4) proves (1.3). Fix a primitive rth root λ. An equivariant injection is injective on the generalized λ-eigenspaces, so both the total dimension μ and the dimension M_r of this individual generalized eigenspace can only increase. At fixed e, r, the balance identity and the fixed value $M_1 = 2 - c$ force equality in both dimensions. The injection is therefore an isomorphism. The argument also applies to semisimplifications. If the contact profiles differ, Theorem 1.1 gives distinct characteristic polynomials, so neither direction admits such an injection. □

Example 6.1 (Equal Milnor numbers and incomparable monodromy)**.** The profiles $\{2, 4, 1^{30}\}$ and $\{3, 3, 1^{30}\}$ both occur with $e = 4, r = 2$, even on the same smooth quartic C. To realize them, choose distinct $p_1, p_2 \in C$ and prescribe $\mathfrak{d} = 2p_1 + 4p_2$ or $3p_1 + 3p_2$. The line bundle $\mathcal{L} = \mathcal{O}_C(9)(-\mathfrak{d})$ has degree 30 on a curve of genus 3. Riemann–Roch gives $h^0(\mathcal{L}) = 28$. Vanishing at either p_i is one independent condition, and a double zero at any other point x imposes two independent conditions, since $H^1(\mathcal{L}(-2x)) = 0$. The double-zero incidence has dimension at most 27 as x varies in C. A general section thus has 30 simple zeros away from p_1, p_2, giving the desired divisor on C.

The restriction of degree-nine homogeneous polynomials to C is surjective, because $H^1(\mathbf{P}^2, \mathcal{O}(5)) = 0$. After lifting the section, replace its lift h_9 by $h_9 + g_4 k_5$. Degree-five polynomials k_5 can adjust the two normal derivatives independently. Choose the resulting derivatives nonzero at both prescribed contacts. At the other intersections, $h_9|_C$ has simple zeros. Thus H is smooth at every intersection and both profiles occur in (1.1).

Their contact polynomials are

$$\Gamma_{2,4}(u) = 1 + 2u + 2u^2 + 2u^3 + u^4, \qquad \Gamma_{3,3}(u) = 1 + 2u + 3u^2 + 2u^3 + u^4.$$

Both have $b = 32$, $M_1 = 6$, and the remaining data are

	μ	M_2	M_{20}	M_{30}
$f_{2,4}$	419	43	2	0
$f_{3,3}$	419	43	0	2

$$\frac{\Delta_{2,4}(t)}{\Delta_{3,3}(t)} = \frac{(1-t^{20})(1-t^{40})}{(1-t^{30})^2}.$$

The two representations have equal dimension and admit no equivariant injection in either direction.

6.2 Contact collisions

Proposition 6.2 (Collision formulas)**.** *Consider two members with the same e, r whose profiles differ by replacing q transverse intersections by one intersection of multiplicity $q \geq 2$. Denote these members by f_{split} and $f_{\tan}$, respectively. Then*

$$\frac{\Delta_{\tan}(t)}{\Delta_{\mathrm{split}}(t)} = \frac{1 - t^{qB}}{1 - t^B} \left(\frac{1 - t^D}{1 - t^{rD}} \right)^{q-1}, \tag{6.1}$$

$$[H^2_{\tan,\mathrm{ss}}] - [H^2_{\mathrm{split},\mathrm{ss}}] = [W_{q,B}] - (q-1)[W_{r,D}], \tag{6.2}$$

$$\mu_{\tan} - \mu_{\mathrm{split}} = q - 1, \qquad M_{r,\tan} - M_{r,\mathrm{split}} = -(q-1). \tag{6.3}$$

The topological functions satisfy

$$Z_{\tan} - Z_{\mathrm{split}} = \frac{(q-1)(r-1)s}{dQ} - \frac{(q-1)rsJ}{dQK_q}. \tag{6.4}$$

Proof. The collision decreases b by $q-1$, adds one factor $(1-t^{qB})/(1-t^B)$, and adds $(q-1)/K_q$ to $U_f(Bs+\beta)$. Substitution into the global formulas proves (6.1), (6.2), and (6.4). Taking dimensions and using $B-(r-1)D=1$ gives (6.3). □

The character difference records the new block $W_{q,B}$ and the decrease of the $W_{r,D}$ contribution. The net gain in dimension is $q-1$, while M_r drops by $q-1$.

Proposition 6.3 (Splitting a single contact)**.** *Suppose that H_0 has one contact of order $q \geq 2$ with C, at p, and all its other intersections with C are transverse. Choose a linear form L with $L(p) \neq 0$ and set*

$$h_{D,\tau} = h_{D,0} + \tau L^D, \qquad f_\tau = g_e^r + h_{D,\tau}. \tag{6.5}$$

For every sufficiently small $\tau \neq 0$, the point p splits into q transverse intersections, all other intersections remain transverse, and f_τ satisfies the assumptions of (1.1).

Proof. Near p use a parameter w on C, with L nonvanishing. The function $\phi = -h_{D,0}|_C / L^D|_C$ has a zero of order q at $w=0$, and ϕ' is w^{q-1} times a unit. Shrink the disc so that 0 is its only critical point. For every sufficiently small nonzero τ, the equation $\phi(w)=\tau$ has q roots by Rouché's theorem, and all are simple because their derivatives do not vanish.

The remaining simple intersections persist by the implicit function theorem. Compactness of C prevents additional zeros outside disjoint neighborhoods of the old intersections. At p the normal derivative of $h_{D,0}$ along C is nonzero, and remains so near the new roots. At all other roots the tangential derivative is nonzero. Thus H_τ is smooth at every intersection, with all the claimed intersection multiplicities. Lemma 2.1 applies to each f_τ. □

6.3 Extremal contact profiles

Proposition 6.4 (Extremal Milnor numbers)**.** *At fixed e, r,*

$$\mu_{\mathrm{tr}} \leq \mu_f \leq \mu_{\mathrm{tr}} + eD - 1, \qquad \mu_{\mathrm{tr}} = A\kappa + (r-1)D(eD-c) - 1. \tag{6.6}$$

Both bounds are attained for every $e \geq 3, r \geq 2$. Equality at the lower bound means that all intersections are transverse; equality at the upper bound means that there is a single intersection of multiplicity eD.

Proof. The classical Milnor formula (4.4) gives $\mu_f - \mu_{\mathrm{tr}} = n - b$, and $1 \leq b \leq n$. For the lower bound take h_D to be a product of D general linear forms. Their intersections with C are transverse and pairwise distinct on C, and H is smooth at each of them.

For the upper bound choose

$$g_e = x^e + y^e + z^e, \qquad \alpha^e = -1, \qquad p = [1 : \alpha : 0], \qquad \ell = x + \alpha^{e-1} y.$$

On $\ell = 0$ one has $g_e = z^e$, so $\mathrm{div}_C(\ell) = e\,p$. Set

$$h_D = \ell^D + g_e x^{D-e}. \tag{6.7}$$

Its restriction to C has divisor $eD\,p$. In the affine chart $x=1$, $dh_D(p) = dg_e(p) \neq 0$, so H is smooth at its unique intersection with C. This realizes the upper bound. □

The maximal-contact member (6.7) has exactly one intersection, so it supplies such a family for every e, r, with $q = eD$.

7 Twisted topological zeta functions

7.1 The divisibility-filtered formula

Fix an integer $\ell > 1$. On a log resolution define

$$Z_f^{(\ell)}(s) = \sum_{\substack{\varnothing \neq I \\ \ell \mid N_i \text{ for every } i \in I}} \chi(E_I^\circ \cap h^{-1}(0)) \prod_{i \in I} (N_i s + \nu_i)^{-1}. \tag{7.1}$$

This is the twisted topological normalization of [2, Section 1.3, equation (1.6)]. Write $\delta(N) = \mathbf{1}_{\ell \mid N}$ and $\eta_q = \gcd(D, q)$.

The auxiliary coordinate boundaries V, W have $N = 0$, so the condition $\ell \mid 0$ preserves the Euler refinement used in the local calculations.

Theorem 7.1. *Set*

$$E_{q,\ell}(s) = q\delta(r) + r\delta(\eta_q) - \delta(r\eta_q) + Q(s)\big[\delta(qB) - q\delta(B)\big]. \tag{7.2}$$

The complete twisted function is

$$Z_f^{(\ell)} = \frac{\kappa\delta(A)}{a} + \frac{c\delta(r)}{aQ} + \frac{(c-b)[r\delta(D) - \delta(rD)]}{Q} + \sum_{q \geq 2} \frac{n_q E_{q,\ell}}{QK_q}. \tag{7.3}$$

Proof. The filter in (7.1) is compatible with the normal crossings change in (2.10). If the old multiplicities at a blowup center are all divisible by ℓ, the new one, their sum, is also divisible, and the same identity applies. If exactly one is not divisible, the new multiplicity is not divisible and all contributions vanish. If at least two are not divisible, every projective stratum of nonzero Euler characteristic retains a nondivisible old component, so again every contribution vanishes. Thus the filtered local functions also assemble by Euler Fubini.

On the regular toric resolution, the strict transform has $N = 1$ and is excluded. The auxiliary boundaries just described are retained for the orbit stratification. Filtering the orbit calculation by the saturated-lattice gcds in Appendix A.3 gives

$$\begin{aligned} Z_{u^A}^{(\ell)} &= \frac{\delta(A)}{a}, \\ Z_G^{(\ell)} &= \frac{\delta(r)}{aQ} + \frac{r\delta(D) - \delta(rD)}{Q}, \\ Z_q^{(\ell)} &= \frac{\delta(r)}{aQ} + \frac{E_{q,\ell}}{QK_q}. \end{aligned} \tag{7.4}$$

When $q = 1$, $\eta_q = 1$ and $E_{1,\ell} = 0$. Apply (2.8) with these local functions; the coefficient of $\delta(r)/(aQ)$ is c, giving (7.3). □

7.2 Holomorphy

Theorem 7.2. *If no monodromy eigenvalue at the origin has order divisible by $\ell > 1$, then $Z_f^{(\ell)}$ is identically zero.*

Proof. By (4.8), every eigenvalue order divides one of

$$A, \qquad rD, \qquad \{qB : q \geq 2,\ n_q > 0\}.$$

Conversely, each of these integers itself occurs as an eigenvalue order: primitive Ath, rDth, and qBth characters have positive multiplicity in the corresponding positive representation blocks. The hypothesis is therefore equivalent to

$$\ell \nmid A, \qquad \ell \nmid rD, \qquad \ell \nmid qB \quad (q \geq 2,\ n_q > 0).$$

It gives $\delta(A) = \delta(rD) = 0$, and the divisibilities $r \mid A$, $D \mid rD$, $\eta_q \mid D$, $r\eta_q \mid rD$ give $\delta(r) = \delta(D) = \delta(\eta_q) = \delta(r\eta_q) = 0$. For each occurring $q \geq 2$ it also gives $\delta(qB) = \delta(B) = 0$. Every term in (7.3) vanishes. □

At smooth points the monodromy has eigenvalue 1, so the same criterion applies to eigenvalues in a neighborhood of the origin.

Example 7.3 (The order condition)**.** For $e = 4, r = 2$ with all intersections transverse, one has $c = -4, b = 36$ and

$$M_3(f) = 0, \qquad Z_f^{(3)}(s) = -\frac{40}{18s + 7}.$$

A primitive 18th root is nevertheless an eigenvalue, and $3 \mid 18$.

Corollary 7.4. *Within the family* (1.1)*, the single polynomial* Δ_f *determines every twisted function* $Z_f^{(\ell)}$*.*

Proof. Recover $e, r, \{q_p\}$ by Theorem 1.1, and use (7.3). □

A Weighted cone calculations

A.1 Orbit contributions and subdivision

We derive the weighted formula of Proposition 3.1. For a nonnegative vector ρ set

$$N(\rho) = \min\{\langle X, \rho\rangle, \langle Y, \rho\rangle, \langle Z, \rho\rangle\}, \qquad \nu(\rho) = 3\rho_u + \rho_v + \rho_w.$$

For a regular toric cone with primitive rays ρ_i, the monomial substitution gives the actual numerical data $N(\rho_i), \nu(\rho_i)$. The reduced strict transform $\mathcal{S}$ has data $(1, 1)$, and the auxiliary boundaries V, W have data $(0, 1)$.

Let σ be a regular cone of dimension k, and let τ be the face selected by its relative interior. Its orbit maps to the origin exactly when $\operatorname{relint}(\sigma)$ contains a vector with all coordinates positive. Thus all relevant selected faces are compact. If $k = 3$, the selected face is a vertex and the orbit is a point disjoint from the strict transform. Its contribution is $\prod_i (N(\rho_i)s + \nu(\rho_i))^{-1}$.

For $k < 3$, the orbit is $(\mathbf{C}^*)^{3-k}$. If $\dim \tau < 3 - k$, the strict-transform equation either has no zero or has an additional torus factor. Both the orbit and its strict-transform zero set have Euler characteristic zero. To see the factor directly, the edge difference vector is primitive and the triangle difference lattice is saturated, as their normalized volumes are one. They can therefore be completed to a basis in the orbit character lattice.

If $\dim \tau = 3 - k \geq 1$, the zero set is one point for an edge, and $1 + \xi + \eta = 0$ in $(\mathbf{C}^*)^2$ for the triangle. Writing $d_\tau = \dim \tau$, its Euler characteristic is $(-1)^{d_\tau - 1}$. The complement and strict-transform terms together contribute

$$\left[-\chi(\mathcal{S} \cap O_\sigma) + \frac{\chi(\mathcal{S} \cap O_\sigma)}{s + 1}\right] \prod_i (N(\rho_i)s + \nu(\rho_i))^{-1} = \frac{s}{s + 1}(-1)^{d_\tau} \prod_i (N(\rho_i)s + \nu(\rho_i))^{-1}.$$

These terms can be merged over nonregular cones while keeping their lattice multiplicities. For a simplicial cone Δ, write

$$J_\Delta(s) = \frac{\operatorname{mult}(\Delta)}{\prod_{\rho_i \text{ ray of } \Delta} (N(\rho_i)s + \nu(\rho_i))}.$$

The multiplicity is the index of its rays in the saturated lattice of its span. On a Newton cone N is linear, and, for real $s > 0$,

$$J_\Delta(s) = \int_\Delta e^{-sN(x) - \nu(x)}\, dx,$$

with the lattice-normalized measure on that span. The integral is additive under subdivisions of the same dimension. Equality of the resulting rational functions then follows by continuation. In particular, J_τ, the sum over a triangulation of the dual cone of τ, is independent of that triangulation. The preceding orbit calculation yields

$$Z_q = J_X + J_Y + J_Z + \frac{s}{s+1}(-J_{XY} - J_{XZ} - J_{YZ} + J_{XYZ}). \tag{A.1}$$

This is the Newton face formula specialized to the pulled-back volume form; compare the standard-volume version in [3, Théorème 5.3].

A.2 The cone table

Using the normals in the proof of Proposition 3.1, triangulate the vertex cones as follows:

Vertex	Simplicial cones	Multiplicities
X	$(\mathsf{U},\mathsf{W},\mathsf{T})$, $(\mathsf{U},\mathsf{R},\mathsf{T})$	$q,\ 1$
Y	$(\mathsf{W},\mathsf{S},\mathsf{T})$	$q^2(r-1)$
Z	$(\mathsf{V},\mathsf{R},\mathsf{T})$, $(\mathsf{V},\mathsf{T},\mathsf{S})$	$r,\ q(r-1)$

Absolute determinants give the last column. Hence

$$J_X = \frac{q}{aK_q} + \frac{1}{aQK_q}, \qquad J_Y = \frac{q^2(r-1)}{p_qK_q}, \qquad J_Z = \frac{r}{QK_q} + \frac{q(r-1)}{p_qK_q}.$$

The two-dimensional lattice indices and the primitive ray T give

$$J_{XY} = \frac{q}{K_q}, \qquad J_{XZ} = \frac{1}{QK_q}, \qquad J_{YZ} = \frac{q(r-1)}{p_qK_q}, \qquad J_{XYZ} = \frac{1}{K_q}.$$

In (A.1), all terms containing p_q combine as

$$\frac{q(r-1)}{p_qK_q}\left(q+1-\frac{s}{s+1}\right) = \frac{q(r-1)}{(s+1)K_q}.$$

Substituting the remaining terms gives (3.6). This identifies the cancellation of the apparent denominator p_q before the global assembly.

A.3 Divisibility on saturated lattices

For a compact face τ, let C_τ be its dual cone and $L_\tau = \operatorname{span}_{\mathbf{R}}(C_\tau) \cap \mathbf{Z}^3$. Choose an exponent $v_\tau \in \tau$ and put

$$m_\tau = \gcd\{\langle v_\tau, \rho\rangle : \rho \in L_\tau\}.$$

The definition is independent of the exponent chosen in the face, since all exponent differences annihilate L_τ. It uses the linear extension of $N|_{C_\tau}$ to this lattice. The primitive rays of a relatively full-dimensional regular subcone of C_τ form a basis of L_τ. Thus all their multiplicities are divisible by ℓ exactly when $\ell \mid m_\tau$, so the divisibility indicator is constant on these subcones. By the lattice-normalized subdivision additivity in Appendix A.1, their filtered cone sum is $\delta(m_\tau)J_\tau$.

For the present triangle these gcds are

τ	X	Y	Z	XY	XZ	YZ	XYZ
m_τ	r	1	η_q	B	$r\eta_q$	η_q	qB

The vertex rows are the gcds of their coordinates. For XY, a lattice basis is $(r-1,1,0),(0,0,1)$, whose pairings are $B,0$. For XZ, the rays R,T are a lattice basis because their cross product is primitive. Thus $m_{XZ}=\gcd(rD,qB)=r\gcd(D,q)$; here $B/r=e(r-1)+1$ is coprime to D. For YZ use $(1,0,0),(0,q,1)$, with pairings D,q. The triangle lattice is generated by T, of multiplicity qB.

For $\ell>1$ the strict-transform terms are excluded, so the orbit signs are $+1$ for vertices, -1 for edges, and $+1$ for the triangle. Since $\delta(1)=0$, the filtered sum is

$$\begin{aligned} Z_q^{(\ell)} &= \delta(r)J_X+\delta(\eta_q)J_Z-\delta(B)J_{XY}-\delta(r\eta_q)J_{XZ} \\ &\quad -\delta(\eta_q)J_{YZ}+\delta(qB)J_{XYZ}. \end{aligned} \tag{A.2}$$

The two p_q terms have the same indicator and cancel. Finally

$$\frac{q}{aK_q}+\frac{1}{aQK_q}=\frac{1}{aQ}+\frac{q}{QK_q}$$

by $K_q=q(Q-a)+1$, giving the $Z_q^{(\ell)}$ formula in (7.4). For G, the two vertex gcds are r,D and the edge gcd is rD; the same two-dimensional calculation gives $\delta(r)/(aQ)+(r\delta(D)-\delta(rD))/Q$.

B An integer-coefficient example

Consider

$$\begin{gathered} g=x^4+y^4+z^4,\qquad p_7=x^7+y^7+z^7,\qquad p_5=x^5+y^5+z^5, \\ h_9=z^2p_7+gp_5,\qquad f=g^2+h_9. \end{gathered} \tag{B.1}$$

Contact profile. On the smooth quartic $C=V(g)$, $\operatorname{div}_C(h_9)=2\operatorname{div}_C(z)+\operatorname{div}_C(p_7)$.

The four points with $z=0$ are $[1:t:0]$, $t^4=-1$, and the line meets C transversely. At these points $p_7=1-t^3\neq 0$ and $p_5=1-t\neq 0$. Thus $h_9|_C$ has order exactly two and $dh_9=p_5\,dg\neq 0$.

The intersection of C and $V(p_7)$ has no point with a zero coordinate. At a nontransverse intersection their gradients are proportional, forcing $x^3=y^3=z^3$. Set $x=1$. The conditions reduce to $y^3=z^3=1$ and $1+y+z=0$, so the only such points are

$$[1:\omega:\omega^2],\qquad [1:\omega^2:\omega],\qquad \omega^2+\omega+1=0.$$

Their contact orders are exactly two. At the first point $\nabla p_7=(7/4)\nabla g$; for the tangent vector $v=(0,1,-1)$,

$$v^{\mathsf{t}}\left(\operatorname{Hess}(p_7)-\frac{7}{4}\operatorname{Hess}(g)\right)v=21(\omega^2+\omega)=-21\neq 0.$$

The second point is identical by symmetry. At both, $p_5=0$ and $dh_9=z^2dp_7\neq 0$.

All remaining intersections of C and $V(p_7)$ are transverse. There is no common component, since $p_7|_C$ is nonzero at the four points with $z=0$. Bézout gives total intersection number 28; after the two double points this leaves 24 transverse points, all with $z\neq 0$. At these points $h_9|_C$ has a simple zero, so H is smooth there as well. Consequently

$$n_2=6,\qquad n_1=24,\qquad b=30,\qquad e=4,\quad r=2,\quad A=8,\quad D=9,\quad B=10.$$

The two invariants. The contact formulas give

$$\begin{aligned} Z_f(s) &= \frac{-3784s^3-1922s^2+321s+189}{(8s+3)(s+1)(18s+7)(20s+9)}, \\ \Delta_f(t) &= \frac{(1-t^8)^7}{1-t}(1+t^9)^{34}(1+t^{10})^6,\qquad \mu_f=421. \end{aligned}$$

The contact pole $-9/20$ has residue $27/1210$. The associated primitive 20th roots have multiplicity six. The balance identity reads $421 + 41 - 6 = 456 = 8^3 - 8^2 + 8$.

Reconstruction. Starting from the characteristic polynomial, the product $P_f = (1-t)\Delta_f$ begins

$$P_f(t) = 1 - 7t^8 + 34t^9 + 6t^{10} + O(t^{16}).$$

Proposition 5.1 gives $A = 8$, $\kappa = 7$, hence $e = 4$, $r = 2$, $c = -4$, and $b = -4 + 34 = 30$. Removing the two determined factors leaves

$$\frac{P_f(t)}{(1-t^8)^7(1+t^9)^{34}} = (1+t^{10})^6 = \Gamma_f(t^{10}).$$

Thus $\Gamma_f(u) = (1+u)^6$. Finite inversion gives $n_2 = 6$ and $n_q = 0$ for $q \geq 3$; finally $n_1 = 30 - 6 = 24$.

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References

[1] N. A'Campo, *La fonction zêta d'une monodromie*, Comment. Math. Helv. **50** (1975), 233–248.

[2] E. Artal Bartolo, P. D. González Pérez, M. González Villa, and E. León Cardenal, *Denef–Loeser zeta functions of suspensions and Lê–Yomdin singularities*, arXiv:2604.24523v1 (2026).

[3] J. Denef and F. Loeser, *Caractéristiques d'Euler–Poincaré, fonctions zêta locales et modifications analytiques*, J. Amer. Math. Soc. **5** (1992), no. 4, 705–720.

[4] S. M. Gusein-Zade, I. Luengo, and A. Melle-Hernández, *Partial resolutions and the zeta-function of a singularity*, Comment. Math. Helv. **72** (1997), no. 2, 244–256.

[5] M. Larson, S. Payne, and A. Stapledon, *The local motivic monodromy conjecture for simplicial nondegenerate singularities*, arXiv:2209.03553v3 (2026).

[6] A. Lemahieu and L. Van Proeyen, *Monodromy conjecture for non-degenerate surface singularities*, Trans. Amer. Math. Soc. **363** (2011), no. 9, 4801–4829.

[7] J. Milnor, *Singular Points of Complex Hypersurfaces*, Annals of Mathematics Studies, vol. 61, Princeton University Press, Princeton, NJ, 1968.

[8] W. Veys, *Introduction to the monodromy conjecture*, arXiv:2403.03343v1 (2024).